

1. Financial Health

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Metric: Non-Revenue Water (NRW) Rate

- **Formula:**
 - Total_billed = Sum of billed for each zone per month across all sources
 - Total_paid = Sum of paid for each zone per month across all sources
 - NRW Rate for each zone per month = $(\text{Total_billed} - \text{Total_paid}) * 100 / \text{Total_billed}$
- **Tooltip:** NRW Amount = Total_billed - Total_paid
- **File:** [Agg_billing](#)

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Metric: Sewer Revenue Coverage (SRC) Rate

- **Formula:**
 - SRC Rate for each city per month = $(\text{sewer_revenue} / \text{opex}) \times 100$
- **Tooltip:** SRC Amount = sewer_revenue
- **Files:** [all_fin_service](#)
- <https://www.mdpi.com/2073-4441/10/1/27#:~:text=WWC%20=%20Number%20of%20W ater%20&%20Wastewater,categories%20defined%20by%20threshold%20values.>

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Opex Share of Budget (OSB) Rate

- **Formula:**
 - Total_opex = For all months of every year, find the sum of opex across each city [all_fin_service](#)
 - Budget_allocated = as it is for every year, from [all_national](#)
 - OSB Rate for each city per year = $\text{Total_opex} * 100 / \text{Budget_allocated}$
- **Tooltips:** Total_opex
- **Files:** [all_fin_service](#) and [all_national](#)

2. Operational Performance

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Metric: Sewer Unresolved Complaints (SUC) Rate

Formula:

- SUC Rate for each city per month = $(\text{complaints} - \text{resolved} / \text{complaints}) \times 100$

- **Tooltip:** SUC = complaints - resolved
- **Files:** [all_fin_service](#)

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Metric: Sewer Blocks per Kilometre (SBK)

- **Formula:**
 - SBK for each city per month = blocks / sewer_length
- **Tooltip:** blocks
- **Files:** [all_fin_service](#)

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Metric: E.Coli Tests Failed (ETF) Rate and E.Coli Tests Skipped (ETS)

- **Formula:**
 - E.Coli Tests Failed for each zone per month = $(\text{test_conducted_ecoli} - \text{tests_passed_ecoli} / \text{test_conducted_ecoli}) \times 100$
 - E.Coli Tests Skipped for each zone per month = $(\text{tests_ecoli} - \text{total_test_conducted_ecoli} / \text{tests_ecoli}) \times 100$
- **Tooltips:** (test_conducted_ecoli-tests_passed_ecoli) and (tests_ecoli-test_conducted_ecoli)
- **Files:** [w_service](#)

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Metric: Chlorine Tests Failed (CTF) Rate and Chlorine Tests Skipped (CTS)

- **Formula:**
 - Chlorine Tests Failed for each zone per month = $(\text{tests_conducted_chlorine} - \text{test_passed_chlorine} / \text{tests_conducted_chlorine}) \times 100$
 - Chlorine Tests Skipped for each zone per month = $(\text{tests_chlorine} - \text{tests_conducted_chlorine} / \text{tests_chlorine}) \times 100$
- **Tooltips:** (tests_conducted_chlorine-test_passed_chlorine) and (tests_chlorine-tests_conducted_chlorine)
- **Files:** [w_service](#)

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Water Staffing per hundred Households (WSH) Rate

Formula:

- Sum the following for all Zones of a city per year: popn_total, households, municipal_coverage **w_access**
- Population per household (PPH) for each city per year = $\text{sum}(\text{popn_total}) / \text{sum}(\text{household})$ **w_access**
- Municipal_households_covered (MHC) for each city per year = $\text{sum}(\text{municipal_coverage}) / \text{PPH}$ **w_access**
- WSH Rate for each city per month = $(\text{w_staff} / \text{MHC of that year}) \times 100$ **all_fin_service**
- **Tooltips:** w_staff
- **Files:** **w_access** and **all_fin_service**

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Sanitation Staffing per 100 Sewer_Connection (SSC) Rate

- **Formula:**
 - city_sewer_connections per month = For every month of each year, find the sum of sewer_connections across all zones of a city **s_service**
 - SSS Rate for each city per month = $(\text{san_staff} / \text{city_sewer_connections}) \times 100$ **all_fin_service**
 - **Tooltips:** san_staff
- **Files:** **s_service** and **all_fin_service**

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Non Consumed Water (NCW) Rate

- **Formula:**
 - Total_production = Sum of all production across all sources on all days of a month **production**
 - Grand_Total_Consumption = Sum of all total_consumption across all zones in a month **w_service**
 - NCW Rate of each country per month = $(\text{Total_production} - \text{Grand_Total_Consumption}) * 100 / \text{Total_production}$
- **Tooltips:** Total_production - Grand_Total_Consumption
- **Files:** **production** and **w_service**

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Non Metered Water (NMW) Rate

Formula:

- $\text{NMW Rate for each zone per month} = (\text{total_consumption} - \text{metered}) * 100 / \text{total_consumption}$
- **Tooltips:** total_consumption - metered
- **Files:** w_service

3. Service Coverage & Equity

Metric Visuals: Water Access (WA) over time (as a stacked bar chart)

File: w_access

Formula (direct columns, no calculation needed) for each zone per year:

safely_managed_pct, basic_pct, limited_pct, unimproved_pct, surface_water_pct

Tooltips: safely_managed, basic, limited, unimproved, surface_water

Metric Visuals: Sanitation Access (SA) over time (as a stacked bar chart)

File: s_access

Formula (direct columns, no calculation needed) for each zone per year:

safely_managed_pct, basic_pct, limited_pct, unimproved_pct, open_def_pct

Tooltips: safely_managed, basic, limited, unimproved, open_def

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Metric: Population Unconnected to Water (PUW) Rate

- **Formula:**
 - $\text{PUW Rate for each zone per year} = (\text{popn_total} - \text{municipal_coverage} / \text{popn_total}) \times 100$
- **Tooltips:** popn_total-municipal_coverage
- **Files:** w_access

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Metric: Households Unconnected to Sanitation (HUS) Rate

- **Formula:**
 - $\text{HUS Rate for each zone per month} = (\text{households} - \text{sewer_connections} / \text{households}) \times 100$
- **Tooltips:** households-sewer_connections
- **Files:** s_service